MECHANICS APPLIED TO AEROSPACE ENGINEERING — PART I

# Topic 5: Relative Motion

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Sources: slides/05_-_Relative_motion.pdf (Slides 1–31) & teoria/Notes.pdf (Sections 1.5 and 1.7, Chapter 3, Section 4.9). Equation numbers are those printed in Notes.pdf.

CENTRAL IDEA OF THE TOPIC

A vector can be *defined* in one reference frame, *differentiated* in another and *expressed* in yet another basis (Slides 17 and 20). Position vectors simply add (Notes Eq. 3.16), but time derivatives depend on the observer. The Coriolis theorem (Notes Eq. 3.8) converts a derivative taken in a moving frame into one taken in a fixed frame, and everything else in the topic (relative velocity, relative acceleration, inertia forces) follows from applying it.

## 1. Introduction: Observers and Reference Frames

### 1.1 Different observers see different motions

When we drive a car or travel on an airplane, our reference frame moves with respect to the frame of an observer on the ground, and the velocities and accelerations we measure differ from those measured on the ground (Notes Ch. 3, p. 36). For a particle $P$ observed from two frames $S_0$ and $S_1$ (Slide 3):

$$\mathbf{r}_0^P \neq \mathbf{r}_1^P, \qquad \mathbf{v}_0^P \neq \mathbf{v}_1^P, \qquad \mathbf{a}_0^P \neq \mathbf{a}_1^P$$

*The subscript names the observer (the frame in which the quantity is measured); the superscript names the point.*

### 1.2 Why we need relative motion

- An **observer** is the same thing as a **reference frame** (Slide 3).
- Quite often one of the two descriptions is easy to compute and the other is the one we need (Notes p. 36).
- Newton's laws apply only in **inertial** frames; any accelerating or rotating frame is **non-inertial**. Observations made in a moving frame must be translated to an inertial frame, or *inertia (fictitious) forces* must be added (Section 11).

### 1.3 Absolute and relative quantities

Quantities measured in the frame called "fixed" are **absolute**; those measured in a "moving" frame are **relative** (Notes p. 36). The naming is a convention: calling a frame "fixed" or "moving", and a position "absolute" or "relative", is completely arbitrary and subjective (Slide 23).

### 1.4 First step: the motion of the frames themselves

To understand relative motion we must first describe how one frame moves with respect to another (Slide 3). A frame $S_1 = \{A; \mathcal{B}_1\}$ has two independent ingredients:

1. **Position and translational motion:** the motion of its origin $A$ (Section 2).
2. **Attitude and rotation:** the orientation of its vector basis $\mathcal{B}_1$ (Sections 3–4) and how it changes in time (Sections 5–8). *Attitude* means orientation.

## 2. Position and Translational Motion of a Reference Frame

Consider $S_0 = \{O; \mathcal{B}_0\}$ and $S_1 = \{A; \mathcal{B}_1\}$. The **translational motion** of $S_1$ with respect to $S_0$ is the motion of the origin $A$ as seen from $S_0$ (Slide 4):

$$\mathbf{r}_0^A, \qquad \mathbf{v}_0^A = \left.\frac{d\mathbf{r}_0^A}{dt}\right|_0, \qquad \mathbf{a}_0^A = \left.\frac{d\mathbf{v}_0^A}{dt}\right|_0$$

Alternatively, the motion of $O$ can be described from $S_1$ with $\mathbf{r}_1^O, \mathbf{v}_1^O, \mathbf{a}_1^O$. Since $\mathbf{r}_1^O = \mathbf{AO} = -\mathbf{OA}$, the positions are opposite, $\mathbf{r}_1^O = -\mathbf{r}_0^A$, but **in general** $\mathbf{v}_1^O \neq -\mathbf{v}_0^A$ and $\mathbf{a}_1^O \neq -\mathbf{a}_0^A$ (Slide 4).

> WHY THE VELOCITIES ARE NOT OPPOSITE
> $\mathbf{v}_1^O$ is the derivative of $\mathbf{r}_1^O$ taken *in* $S_1$, while $\mathbf{v}_0^A$ is the derivative of $\mathbf{r}_0^A$ taken *in* $S_0$. If $\mathcal{B}_1$ rotates, the two derivatives of the same vector differ. The Coriolis theorem of Section 6 gives $\mathbf{v}_1^O = -\left.\frac{d\mathbf{r}_0^A}{dt}\right|_1 = -\left(\mathbf{v}_0^A - \boldsymbol{\omega}_{10} \times \mathbf{r}_0^A\right)$, which equals $-\mathbf{v}_0^A$ only when $\boldsymbol{\omega}_{10} \times \mathbf{r}_0^A = \mathbf{0}$ (for instance, when $S_1$ does not rotate).

## 3. Attitude of a Reference Frame: the Change of Basis

### 3.1 Definition of attitude

The **attitude** of $S_1$ with respect to $S_0$ is its instantaneous orientation in space. The position of the origin does not matter: only the directions of the vectors of the bases (Slide 5).

### 3.2 Change-of-basis relations

The simplest description writes the vectors of $\mathcal{B}_1$ in terms of those of $\mathcal{B}_0$ (Slide 5, Notes Eq. 1.8):

$$\begin{cases} \mathbf{i}_1 = \ell_{x1,x0}\,\mathbf{i}_0 + \ell_{x1,y0}\,\mathbf{j}_0 + \ell_{x1,z0}\,\mathbf{k}_0 \\ \mathbf{j}_1 = \ell_{y1,x0}\,\mathbf{i}_0 + \ell_{y1,y0}\,\mathbf{j}_0 + \ell_{y1,z0}\,\mathbf{k}_0 \\ \mathbf{k}_1 = \ell_{z1,x0}\,\mathbf{i}_0 + \ell_{z1,y0}\,\mathbf{j}_0 + \ell_{z1,z0}\,\mathbf{k}_0 \end{cases}$$

$\ell_{x1,x0} = \mathbf{i}_1 \cdot \mathbf{i}_0$ *is the cosine of the angle between two unit vectors (a direction cosine).*

### 3.3 Components of a vector in either basis

Any vector $\mathbf{b}$ can be written in either basis; the vector is the same physical entity (Slide 5, Notes Eq. 1.6):

$$\mathbf{b} = b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0 = b_{x1}\mathbf{i}_1 + b_{y1}\mathbf{j}_1 + b_{z1}\mathbf{k}_1$$

**Derivation (Notes Eqs. 1.9–1.12).** Substitute the change-of-basis relations into the second expression:

$$\mathbf{b} = b_{x1}\left(\ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0\right) + b_{y1}\left(\ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0\right) + b_{z1}\left(\ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0\right)$$

Collect the coefficients of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ and compare with $b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0$ (components in a basis are unique):

$$\begin{aligned} b_{x0} &= \ell_{x1,x0}b_{x1} + \ell_{y1,x0}b_{y1} + \ell_{z1,x0}b_{z1} \\ b_{y0} &= \ell_{x1,y0}b_{x1} + \ell_{y1,y0}b_{y1} + \ell_{z1,y0}b_{z1} \\ b_{z0} &= \ell_{x1,z0}b_{x1} + \ell_{y1,z0}b_{y1} + \ell_{z1,z0}b_{z1} \end{aligned}$$

In matrix form (Slide 5, Notes Eqs. 1.12–1.13):

$$\begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix}_0 = [{}_0R_1] \cdot \begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix}_1, \qquad [{}_0R_1] = \begin{bmatrix} \ell_{x1,x0} & \ell_{y1,x0} & \ell_{z1,x0} \\ \ell_{x1,y0} & \ell_{y1,y0} & \ell_{z1,y0} \\ \ell_{x1,z0} & \ell_{y1,z0} & \ell_{z1,z0} \end{bmatrix}$$

$[{}_0R_1]$ *is the change-of-basis matrix from* $\mathcal{B}_1$ *to* $\mathcal{B}_0$*. The subscript on each column matrix names its basis; without it the expression is ambiguous.*

> **CHANGING BASIS DOES NOT MOVE THE VECTOR**
> When the components of $\mathbf{b}$ change from one basis to another, the vector is not moved and not rotated: it is only described from a different basis, so the numbers that represent it change (Notes Sec. 1.5).

### 3.4 Properties of $[{}_0R_1]$

For two right-handed orthonormal bases (Slide 6, Notes Eqs. 1.14–1.15):

1. **Orthogonal:** the inverse equals the transpose, $[{}_1R_0] = [{}_0R_1]^{-1} = [{}_0R_1]^T$, so $[{}_0R_1]^T[{}_0R_1] = [U]$. Writing $\mathbf{i}_0 = \ell_{x0,x1}\mathbf{i}_1 + \ell_{x0,y1}\mathbf{j}_1 + \ell_{x0,z1}\mathbf{k}_1$, etc., this means $\ell_{x0,x1} = \ell_{x1,x0}$, $\ell_{x0,y1} = \ell_{y1,x0}$, and so on; consistent with $\mathbf{i}_0 \cdot \mathbf{j}_1 = \mathbf{j}_1 \cdot \mathbf{i}_0$.
2. **Unit determinant:** $|[{}_0R_1]| = 1$, because both bases are right-handed.
3. **Columns** = components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$. This is the practical way to write the matrix.
4. **Rows** = components of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ in $\mathcal{B}_1$.
5. **Eigenvalues:** $\lambda = 1,\ e^{i\beta},\ e^{-i\beta}$.

### 3.5 Axis and angle of rotation

A result due to Euler states that any two right-handed bases are related by a **single rotation** about some axis $\mathbf{e}$ by some angle (Notes Sec. 1.5.1). In terms of the matrix (Slide 6): the **axis of rotation** is the eigenspace of $\lambda = 1$ (a vector on it has identical components in $\mathcal{B}_1$ and $\mathcal{B}_0$, since $[{}_0R_1]\mathbf{e} = \mathbf{e}$), and the **rotation angle** is $\beta$. The same matrix can be read as the static relation between components in two bases, or as the rotation that, seen from $\mathcal{B}_0$, carries $\mathcal{B}_0$ onto $\mathcal{B}_1$; for this reason it is called a **rotation matrix** (Notes Sec. 1.5.1).

### 3.6 Composition rules

With three bases $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2$, substituting the expressions of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ into $\mathbf{i}_2 = \ell_{x2,x1}\mathbf{i}_1 + \ell_{x2,y1}\mathbf{j}_1 + \ell_{x2,z1}\mathbf{k}_1$ gives $\mathbf{i}_2$ directly in $\mathcal{B}_0$ (Slide 7):

$$\mathbf{i}_2 = \ell_{x2,x1}\left(\ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0\right) + \ell_{x2,y1}\left(\ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0\right) + \ell_{x2,z1}\left(\ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0\right)$$

Equivalently, from $[\mathbf{b}]_0 = [{}_0R_1][\mathbf{b}]_1$ and $[\mathbf{b}]_1 = [{}_1R_2][\mathbf{b}]_2$ (Notes Eq. 1.16):

$$[\mathbf{b}]_0 = [{}_0R_1] \cdot [{}_1R_2] \cdot [\mathbf{b}]_2 \quad \Longrightarrow \quad [{}_0R_2] = [{}_0R_1] \cdot [{}_1R_2]$$

*The product is in general not commutative:* $[R_i][R_j] \neq [R_j][R_i]$.

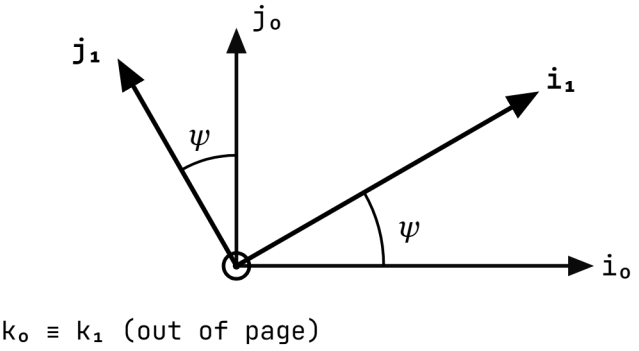
## 4. Simple Rotations and Euler Angles

### 4.1 Simple rotations

A **simple rotation** has its axis along one of the basis vectors. That vector is unchanged (e.g. $\mathbf{i}_1 \equiv \mathbf{i}_0$) and the rest of space turns by an angle that is positive by the right-hand rule (Slide 8, Notes Sec. 1.5.2). Writing the new vectors in the old ones and placing them as columns (Notes Eqs. 1.18–1.20):

| Axis | Relations | Matrix |
|---|---|---|
| $\mathbf{i}_0$, angle $\theta_x$ | $\mathbf{i}_1 = \mathbf{i}_0$<br>$\mathbf{j}_1 = \cos\theta_x\,\mathbf{j}_0 + \sin\theta_x\,\mathbf{k}_0$<br>$\mathbf{k}_1 = -\sin\theta_x\,\mathbf{j}_0 + \cos\theta_x\,\mathbf{k}_0$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{bmatrix}$ |
| $\mathbf{j}_0$, angle $\theta_y$ | $\mathbf{i}_1 = \cos\theta_y\,\mathbf{i}_0 - \sin\theta_y\,\mathbf{k}_0$<br>$\mathbf{j}_1 = \mathbf{j}_0$<br>$\mathbf{k}_1 = \sin\theta_y\,\mathbf{i}_0 + \cos\theta_y\,\mathbf{k}_0$ | $\begin{bmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$ |
| $\mathbf{k}_0$, angle $\theta_z$ | $\mathbf{i}_1 = \cos\theta_z\,\mathbf{i}_0 + \sin\theta_z\,\mathbf{j}_0$<br>$\mathbf{j}_1 = -\sin\theta_z\,\mathbf{i}_0 + \cos\theta_z\,\mathbf{j}_0$<br>$\mathbf{k}_1 = \mathbf{k}_0$ | $\begin{bmatrix} \cos\theta_z & -\sin\theta_z & 0 \\ \sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |

**Check** (for $z$): $\det = \cos^2\theta_z + \sin^2\theta_z = 1$, and the first two columns are orthogonal: $(\cos\theta_z)(-\sin\theta_z) + (\sin\theta_z)(\cos\theta_z) + 0 = 0$. The inverse matrices are the same with $\theta \to -\theta$: only the position of the minus sign before the sines changes. To avoid sign errors, always draw the rotation looking down from the stationary axis (Notes Sec. 1.5.2). The $y$ case looks different only because, seen from $\mathbf{j}_0$, the plane is $Ozx$ in that cyclic order.




*Simple rotation about $\mathbf{k}_0$ by $\psi$ (Slide 16), drawn looking down from the stationary axis as recommended in Notes Sec. 1.5.2.*

## 4.2 Euler angles

Any attitude is the composition of **three simple rotations about different axes**. To relate $\mathcal{B}_0$ to $\mathcal{B}_3$ we introduce the auxiliary bases $\mathcal{B}_1$ and $\mathcal{B}_2$ (Slide 9):

1. $\mathcal{B}_0 \to \mathcal{B}_1$: rotation $\psi$ about $\mathbf{k}_0 \equiv \mathbf{k}_1$;
2. $\mathcal{B}_1 \to \mathcal{B}_2$: rotation $\theta$ about $\mathbf{i}_1 \equiv \mathbf{i}_2$;
3. $\mathcal{B}_2 \to \mathcal{B}_3$: rotation $\varphi$ about $\mathbf{k}_2 \equiv \mathbf{k}_3$.

$$[{}_0R_3] = [{}_0R_1] \cdot [{}_1R_2] \cdot [{}_2R_3]$$

$\psi$ (**precession**), $\theta$ (**nutation**) and $\varphi$ (**spin**) are the **standard Euler angles**, also called **3-1-3** angles after the order of the axes: third, first, third again. Many other choices exist; a common one is **3-2-1 (yaw, pitch, roll)**. The only rule is that the same axis cannot be repeated in two consecutive rotations (Slide 9). The Notes also use the 1-2-3 sequence (Notes Eq. 1.23).

### 4.3 Example: 3-2-1 attitude of the Space Shuttle

The attitude of the red frame $S_r$ (attached to the shuttle) relative to the blue frame $S_b$ is described with 3-2-1 angles, the convention of flight mechanics; both frames start coincident (Slides 10–13).

1. **First rotation**, $\psi$ about $\mathbf{k}_b \equiv \mathbf{k}_1$ (Slide 11): $\mathbf{i}_1 = \cos\psi\,\mathbf{i}_b + \sin\psi\,\mathbf{j}_b,\ \mathbf{j}_1 = -\sin\psi\,\mathbf{i}_b + \cos\psi\,\mathbf{j}_b,\ \mathbf{k}_1 = \mathbf{k}_b$.
2. **Second rotation**, $\theta$ about $\mathbf{j}_1 \equiv \mathbf{j}_2$ (Slide 12): $\mathbf{i}_2 = \cos\theta\,\mathbf{i}_1 - \sin\theta\,\mathbf{k}_1,\ \mathbf{j}_2 = \mathbf{j}_1,\ \mathbf{k}_2 = \sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1$.
3. **Third rotation**, $\varphi$ about $\mathbf{i}_2 \equiv \mathbf{i}_r$ (Slide 13): $\mathbf{i}_r = \mathbf{i}_2,\ \mathbf{j}_r = \cos\varphi\,\mathbf{j}_2 + \sin\varphi\,\mathbf{k}_2,\ \mathbf{k}_r = -\sin\varphi\,\mathbf{j}_2 + \cos\varphi\,\mathbf{k}_2$.

Composing the three matrices, $[{}_bR_r] = [{}_bR_1(z,\psi)] \cdot [{}_1R_2(y,\theta)] \cdot [{}_2R_r(x,\varphi)]$:

$$[{}_bR_r] = \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta\sin\varphi - \sin\psi\cos\varphi & \cos\psi\sin\theta\cos\varphi + \sin\psi\sin\varphi \\ \sin\psi\cos\theta & \sin\psi\sin\theta\sin\varphi + \cos\psi\cos\varphi & \sin\psi\sin\theta\cos\varphi - \cos\psi\sin\varphi \\ -\sin\theta & \cos\theta\sin\varphi & \cos\theta\cos\varphi \end{bmatrix}$$

*First column:* $\mathbf{i}_r = \mathbf{i}_2 = \cos\theta(\cos\psi\,\mathbf{i}_b + \sin\psi\,\mathbf{j}_b) - \sin\theta\,\mathbf{k}_b$*. Check: its squared norm is* $\cos^2\theta(\cos^2\psi + \sin^2\psi) + \sin^2\theta = 1$*.*

## 5. Angular Motion of a Reference Frame

### 5.1 Angular velocity vector

So far the change of basis was stationary, but bases can rotate with respect to each other in time. The rotation of $\mathcal{B}_1$ relative to $\mathcal{B}_0$ is represented at each instant by the **angular velocity vector** $\boldsymbol{\omega}_{10}$ (Slide 14):

- its **direction** is the instantaneous axis of rotation of $\mathcal{B}_1$ (right-hand rule);
- its **magnitude** is the angular rate.

$\boldsymbol{\omega}_{10}$ and the motion of the origin of $S_1$ are unrelated: two completely different things (Slide 14).

**WHERE $\boldsymbol{\omega}$ COMES FROM**

A vector $\mathbf{A}$ of constant magnitude rotating about an axis $\mathbf{e}_\omega$ at rate $\dot\theta$ keeps its component along $\mathbf{e}_\omega$, while its perpendicular part describes a circle. In a time $dt$ it changes by $|d\mathbf{A}| = A_\perp\, d\theta = \dot\theta\,|\mathbf{e}_\omega \times \mathbf{A}|\, dt$, in the direction of $\mathbf{e}_\omega \times \mathbf{A}$ (Notes Sec. 1.7.1). Hence $\dfrac{d\mathbf{A}}{dt} = \dot\theta\,\mathbf{e}_\omega \times \mathbf{A} = \boldsymbol{\omega} \times \mathbf{A}$ with $\boldsymbol{\omega} = \dot\theta\,\mathbf{e}_\omega$ (Notes Eq. 1.62).

## 5.2 Basis vectors of $\mathcal{B}_1$ seen from $S_0$

The vectors of $\mathcal{B}_1$ are unit vectors, so they change only in direction (Slide 14, Notes Eqs. 3.3–3.5):

$$\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{i}_1, \qquad \left.\frac{d\mathbf{j}_1}{dt}\right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{j}_1, \qquad \left.\frac{d\mathbf{k}_1}{dt}\right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{k}_1$$

## 5.3 A vector that is constant in $S_1$

Let $\mathbf{c} = c_{x1}\mathbf{i}_1 + c_{y1}\mathbf{j}_1 + c_{z1}\mathbf{k}_1$ with constant components. In $S_0$, the product rule gives $\dot{c}_{x1}\mathbf{i}_1 + c_{x1}\, d\mathbf{i}_1/dt|_0$ for each term, with $\dot{c}_{x1} = 0$ (Slide 14):

$$\begin{aligned}\left.\frac{d\mathbf{c}}{dt}\right|_0 &= c_{x1}\left.\frac{d\mathbf{i}_1}{dt}\right|_0 + c_{y1}\left.\frac{d\mathbf{j}_1}{dt}\right|_0 + c_{z1}\left.\frac{d\mathbf{k}_1}{dt}\right|_0 = c_{x1}\,\boldsymbol{\omega}_{10} \times \mathbf{i}_1 + c_{y1}\,\boldsymbol{\omega}_{10} \times \mathbf{j}_1 + c_{z1}\,\boldsymbol{\omega}_{10} \times \mathbf{k}_1 \\ &= \boldsymbol{\omega}_{10} \times (c_{x1}\mathbf{i}_1 + c_{y1}\mathbf{j}_1 + c_{z1}\mathbf{k}_1) = \boldsymbol{\omega}_{10} \times \mathbf{c}\end{aligned}$$

*Scalar factors move inside the cross product and the cross product is distributive.*

## 5.4 Angular acceleration vector

The rotation can be accelerated. The **angular acceleration vector** is (Slide 15)

$$\boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0$$

- Its direction may differ from that of $\boldsymbol{\omega}_{10}$.
- $\boldsymbol{\omega}_{10}$ does not describe the current attitude of $S_1$, only how it is changing in time.
- The component of $\boldsymbol{\alpha}_{10}$ **parallel** to $\boldsymbol{\omega}_{10}$ describes how the **magnitude** of $\boldsymbol{\omega}_{10}$ changes; the **perpendicular** component describes how its **direction** changes. With $\boldsymbol{\omega}_{10} = \omega_{10}\mathbf{e}_\omega$, the product rule gives $\boldsymbol{\alpha}_{10} = \dot{\omega}_{10}\,\mathbf{e}_\omega + \omega_{10}\, d\mathbf{e}_\omega/dt|_0$, and $d\mathbf{e}_\omega/dt \perp \mathbf{e}_\omega$ because $\mathbf{e}_\omega$ has unit length (Notes Eq. 1.53).

### 5.5 Example: simple rotation about $\mathbf{k}_0$ (Slide 16)

At $t = 0$ the frames coincide; then $S_1$ rotates about $\mathbf{k}_0$ by $\psi = \psi(t)$: $\mathbf{i}_1 = \cos\psi\,\mathbf{i}_0 + \sin\psi\,\mathbf{j}_0$, $\mathbf{j}_1 = -\sin\psi\,\mathbf{i}_0 + \cos\psi\,\mathbf{j}_0$, $\mathbf{k}_1 = \mathbf{k}_0$.

**Direct differentiation in $S_0$** ($\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ are constant there). Chain rule: $\frac{d}{dt}\cos\psi(t) = -\sin\psi\,\dot\psi$, $\frac{d}{dt}\sin\psi(t) = \cos\psi\,\dot\psi$:

$$\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = \dot\psi\,(-\sin\psi\,\mathbf{i}_0 + \cos\psi\,\mathbf{j}_0) = \dot\psi\,\mathbf{j}_1, \qquad \left.\frac{d\mathbf{j}_1}{dt}\right|_0 = \dot\psi\,(-\cos\psi\,\mathbf{i}_0 - \sin\psi\,\mathbf{j}_0) = -\dot\psi\,\mathbf{i}_1, \qquad \left.\frac{d\mathbf{k}_1}{dt}\right|_0 = \mathbf{0}$$

**Identification.** With $\boldsymbol{\omega}_{10} = \dot\psi\,\mathbf{k}_1$: $\dot\psi\,\mathbf{k}_1 \times \mathbf{i}_1 = \dot\psi\,\mathbf{j}_1$, $\dot\psi\,\mathbf{k}_1 \times \mathbf{j}_1 = -\dot\psi\,\mathbf{i}_1$, $\dot\psi\,\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, which reproduce the three derivatives. Therefore

$$\boldsymbol{\omega}_{10} = \dot\psi\,\mathbf{k}_0, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d(\dot\psi\,\mathbf{k}_0)}{dt}\right|_0 = \ddot\psi\,\mathbf{k}_0$$

*Here $\boldsymbol{\alpha}_{10} \parallel \boldsymbol{\omega}_{10}$: only the magnitude of the angular velocity changes.*

## 6. The Coriolis Theorem

WHY THIS THEOREM?

Positions, velocities and forces are often easiest to write in the basis of a moving frame (the wire, the ring, the aircraft). Newton's law, however, needs derivatives taken in the inertial frame. The Coriolis theorem is the bridge: it gives the derivative in $S_0$ of a vector whose components are known in $\mathcal{B}_1$ (Slide 17, Notes Sec. 3.1).

### 6.1 Derivation

Consider $\mathbf{b} = b_{x1}(t)\,\mathbf{i}_1 + b_{y1}(t)\,\mathbf{j}_1 + b_{z1}(t)\,\mathbf{k}_1$, not constant in $S_1$ (Slide 17; the Notes call it $\mathbf{A}$, Notes Eq. 3.1).

**Derivative in $S_1$.** In $S_1$ the basis vectors are constant, so only the components change (Notes Eq. 3.7):

$$\left.\frac{d\mathbf{b}}{dt}\right|_1 = \frac{db_{x1}}{dt}\mathbf{i}_1 + \frac{db_{y1}}{dt}\mathbf{j}_1 + \frac{db_{z1}}{dt}\mathbf{k}_1$$

**Derivative in $S_0$.** The basis vectors of $\mathcal{B}_1$ are not constant in $S_0$, so the product rule applies to each term, $\frac{d}{dt}(b_{x1}\mathbf{i}_1) = \dot b_{x1}\mathbf{i}_1 + b_{x1}\left.\frac{d\mathbf{i}_1}{dt}\right|_0$ (Notes Eq. 3.2):

$$\left.\frac{d\mathbf{b}}{dt}\right|_0 = \underbrace{\frac{db_{x1}}{dt}\mathbf{i}_1 + \frac{db_{y1}}{dt}\mathbf{j}_1 + \frac{db_{z1}}{dt}\mathbf{k}_1}_{d\mathbf{b}/dt|_1} + \underbrace{b_{x1}\left.\frac{d\mathbf{i}_1}{dt}\right|_0 + b_{y1}\left.\frac{d\mathbf{j}_1}{dt}\right|_0 + b_{z1}\left.\frac{d\mathbf{k}_1}{dt}\right|_0}_{\boldsymbol{\omega}_{10}\times\mathbf{b}}$$

*The second group is the computation of Section 5.3 with the current values of the components (Notes Eq. 3.6).*

$$\boxed{\left.\frac{d\mathbf{b}}{dt}\right|_0 = \left.\frac{d\mathbf{b}}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\mathbf{b}}$$

*Coriolis theorem (Notes Eq. 3.8, Slide 17).*

## 6.2 Meaning and remarks

- The derivative of $\mathbf{b}$ in $S_0$ equals its derivative in $S_1$ plus the rotation of $\mathbf{b}$ due to the angular motion of $S_1$ relative to $S_0$ (Slide 17).
- For vectors, the frame of each derivative must always be written. Derivatives of **scalars** are the same in every frame (Notes p. 37).
- The result holds unchanged when $S_1$ also translates: a translation does not change the orientation of $\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ (Notes p. 37).
- **Remember:** a vector can be defined in one frame, differentiated in another and expressed in yet another (Slide 17).

## 6.3 Consequence: $\boldsymbol{\alpha}_{10}$ is the same in $S_0$ and $S_1$

$$\boldsymbol{\alpha}_{10} \equiv \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_1 + \underbrace{\boldsymbol{\omega}_{10}\times\boldsymbol{\omega}_{10}}_{=\mathbf{0}} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_1$$

*Notes Eq. 3.9. The angular acceleration is in general not parallel to $\boldsymbol{\omega}_{10}$.*

**Example (Slide 18): derivative of $\mathbf{b} = t\,\mathbf{i}_1 + t^2\,\mathbf{j}_1 + t^3\,\mathbf{k}_1$** Slide 18

$S_1$ rotates about $\mathbf{k}_0 \equiv \mathbf{k}_1$ by $\psi(t)$, so $\boldsymbol{\omega}_{10} = \dot{\psi}\,\mathbf{k}_1$

---

1. **In $S_1$:** $\frac{d}{dt}t = 1$, $\frac{d}{dt}t^2 = 2t$, $\frac{d}{dt}t^3 = 3t^2$, so $\left.\frac{d\mathbf{b}}{dt}\right|_1 = \mathbf{i}_1 + 2t\,\mathbf{j}_1 + 3t^2\,\mathbf{k}_1$.
2. **Rotation term**, with $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$: $\boldsymbol{\omega}_{10}\times\mathbf{b} = \dot{\psi}\,t\,\mathbf{j}_1 - \dot{\psi}\,t^2\,\mathbf{i}_1$.
3. **In $S_0$:** $\left.\frac{d\mathbf{b}}{dt}\right|_0 = \left(1 - \dot{\psi}\,t^2\right)\mathbf{i}_1 + \left(2t + \dot{\psi}\,t\right)\mathbf{j}_1 + 3t^2\,\mathbf{k}_1$. This derivative *taken in* $S_0$ is *expressed in* $\mathcal{B}_1$.

4. **Cross-check**: writing $\mathbf{b}$ in $\mathcal{B}_0$ with the relations of Section 5.5 and differentiating component by component gives the same vector (checked symbolically).

# 7. Composition of Angular Motions

## 7.1 Angular velocities add

If $S_2$ rotates relative to $S_1$ with $\boldsymbol{\omega}_{21}$ and $S_1$ relative to $S_0$ with $\boldsymbol{\omega}_{10}$ (Slide 19, Notes Eq. 3.10):

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10}$$

*For any $\mathbf{c}$ constant in $S_2$: $\left.\frac{d\mathbf{c}}{dt}\right|_1 = \boldsymbol{\omega}_{21} \times \mathbf{c}$, and by Coriolis $\left.\frac{d\mathbf{c}}{dt}\right|_0 = \boldsymbol{\omega}_{21} \times \mathbf{c} + \boldsymbol{\omega}_{10} \times \mathbf{c}$, which must equal $\boldsymbol{\omega}_{20} \times \mathbf{c}$ for every $\mathbf{c}$.*

## 7.2 Angular accelerations do not simply add

Differentiate in $S_0$ (Slide 19, Notes Eq. 3.11). The last term is $\boldsymbol{\alpha}_{10}$; the first is the $S_0$-derivative of a vector naturally defined in $S_1$, so by Coriolis $\left.\frac{d\boldsymbol{\omega}_{21}}{dt}\right|_0 = \left.\frac{d\boldsymbol{\omega}_{21}}{dt}\right|_1 + \boldsymbol{\omega}_{10} \times \boldsymbol{\omega}_{21}$:

$$\left.\frac{d\boldsymbol{\omega}_{20}}{dt}\right|_0 = \left.\frac{d\boldsymbol{\omega}_{21}}{dt}\right|_0 + \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 \quad \Longrightarrow \quad \boldsymbol{\alpha}_{20} = \boldsymbol{\alpha}_{21} + \boldsymbol{\omega}_{10} \times \boldsymbol{\omega}_{21} + \boldsymbol{\alpha}_{10}$$

*Notes Eq. 3.12. The extra term appears because $\boldsymbol{\omega}_{21}$ is carried around by the rotation of $S_1$.*

## 7.3 Inverse motion

$$\boldsymbol{\omega}_{01} = -\boldsymbol{\omega}_{10}, \qquad \boldsymbol{\alpha}_{01} = -\boldsymbol{\alpha}_{10}$$

*Slide 19, Notes Eq. 3.13: take $S_2 = S_0$ in Section 7.1, $\boldsymbol{\omega}_{00} = \mathbf{0} = \boldsymbol{\omega}_{01} + \boldsymbol{\omega}_{10}$, then differentiate.*

In general $\boldsymbol{\omega}_{lm}$ is the angular velocity of frame $l$ as seen from frame $m$. When composing in components, **express every vector in the same basis first** (Notes p. 38).

# 8. Angular Velocity from Simple Rotations

## 8.1 3-1-3 Euler angles as functions of time

For $\psi(t), \theta(t), \varphi(t)$, $S_0 \to S_1 \to S_2 \to S_3$ (Slide 20). Each simple rotation has the angular velocity "rate of its angle times its axis" (Section 5.5):

$$\boldsymbol{\omega}_{10} = \dot{\psi}\,\mathbf{k}_0 \equiv \dot{\psi}\,\mathbf{k}_1, \qquad \boldsymbol{\omega}_{21} = \dot{\theta}\,\mathbf{i}_1 \equiv \dot{\theta}\,\mathbf{i}_2, \qquad \boldsymbol{\omega}_{32} = \dot{\varphi}\,\mathbf{k}_2 \equiv \dot{\varphi}\,\mathbf{k}_3$$

$$\boldsymbol{\omega}_{30} = \boldsymbol{\omega}_{32} + \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10} = \dot{\varphi}\,\mathbf{k}_2 + \dot{\theta}\,\mathbf{i}_1 + \dot{\psi}\,\mathbf{k}_0$$

*A mixed expression: each term is written in a different basis (Slide 20).*

## 8.2 Expressed in $\mathcal{B}_0$ (Slide 21)

From $\mathbf{j}_1 = -\sin\psi\,\mathbf{i}_0 + \cos\psi\,\mathbf{j}_0$ and $\mathbf{k}_2 = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$:

$$\mathbf{k}_2 = \sin\theta\sin\psi\,\mathbf{i}_0 - \sin\theta\cos\psi\,\mathbf{j}_0 + \cos\theta\,\mathbf{k}_0, \qquad \mathbf{i}_1 = \cos\psi\,\mathbf{i}_0 + \sin\psi\,\mathbf{j}_0$$

$$\boldsymbol{\omega}_{30} = \Big(\dot{\theta}\cos\psi + \dot{\varphi}\sin\psi\sin\theta\Big)\,\mathbf{i}_0 + \Big(\dot{\theta}\sin\psi - \dot{\varphi}\cos\psi\sin\theta\Big)\,\mathbf{j}_0 + \Big(\dot{\varphi}\cos\theta + \dot{\psi}\Big)\,\mathbf{k}_0$$

## 8.3 Expressed in $\mathcal{B}_3$ (Slide 21)

Inverting the last rotation, $\mathbf{i}_2 = \cos\varphi\,\mathbf{i}_3 - \sin\varphi\,\mathbf{j}_3$, $\mathbf{j}_2 = \sin\varphi\,\mathbf{i}_3 + \cos\varphi\,\mathbf{j}_3$; inverting the second, $\mathbf{k}_1 = \sin\theta\,\mathbf{j}_2 + \cos\theta\,\mathbf{k}_2$. Then

$$\mathbf{i}_1 = \cos\varphi\,\mathbf{i}_3 - \sin\varphi\,\mathbf{j}_3, \qquad \mathbf{k}_0 = \sin\theta\sin\varphi\,\mathbf{i}_3 + \sin\theta\cos\varphi\,\mathbf{j}_3 + \cos\theta\,\mathbf{k}_3$$

$$\boldsymbol{\omega}_{30} = \Big(\dot{\theta}\cos\varphi + \dot{\psi}\sin\varphi\sin\theta\Big)\,\mathbf{i}_3 + \Big(-\dot{\theta}\sin\varphi + \dot{\psi}\cos\varphi\sin\theta\Big)\,\mathbf{j}_3 + \Big(\dot{\varphi} + \dot{\psi}\cos\theta\Big)\,\mathbf{k}_3$$

### 8.4 Exercise of Slide 21: 3-2-1 (yaw-pitch-roll) angles

With the rotations of Section 4.3: $\boldsymbol{\omega}_{rb} = \dot{\psi}\,\mathbf{k}_b + \dot{\theta}\,\mathbf{j}_1 + \dot{\varphi}\,\mathbf{i}_2$.

**In $\mathcal{B}_b$**, with $\mathbf{j}_1 = -\sin\psi\,\mathbf{i}_b + \cos\psi\,\mathbf{j}_b$ and $\mathbf{i}_2 = \cos\theta\cos\psi\,\mathbf{i}_b + \cos\theta\sin\psi\,\mathbf{j}_b - \sin\theta\,\mathbf{k}_b$:

$$\boldsymbol{\omega}_{rb} = \Big(\dot{\varphi}\cos\theta\cos\psi - \dot{\theta}\sin\psi\Big)\,\mathbf{i}_b + \Big(\dot{\varphi}\cos\theta\sin\psi + \dot{\theta}\cos\psi\Big)\,\mathbf{j}_b + \Big(\dot{\psi} - \dot{\varphi}\sin\theta\Big)\,\mathbf{k}_b$$

**In $\mathcal{B}_r$:** $\mathbf{j}_1 = \mathbf{j}_2 = \cos\varphi\,\mathbf{j}_r - \sin\varphi\,\mathbf{k}_r$ and $\mathbf{k}_b = \mathbf{k}_1 = -\sin\theta\,\mathbf{i}_r + \cos\theta\sin\varphi\,\mathbf{j}_r + \cos\theta\cos\varphi\,\mathbf{k}_r$ (inverting the second and third rotations), so

$$\boldsymbol{\omega}_{rb} = \Big(\dot{\varphi} - \dot{\psi}\sin\theta\Big)\,\mathbf{i}_r + \Big(\dot{\theta}\cos\varphi + \dot{\psi}\cos\theta\sin\varphi\Big)\,\mathbf{j}_r + \Big(\dot{\psi}\cos\theta\cos\varphi - \dot{\theta}\sin\varphi\Big)\,\mathbf{k}_r$$

*Both 3-1-3 and 3-2-1 results were checked symbolically against $[\Omega] = [\dot{R}]\,[R]^T$ (Notes Sec. 1.7.2).*

## 9. Configuration and Motion of a Reference Frame

The configuration and motion of $S_1 = \{A; \mathcal{B}_1\}$ relative to $S_0$ are completely specified by (Slide 22):

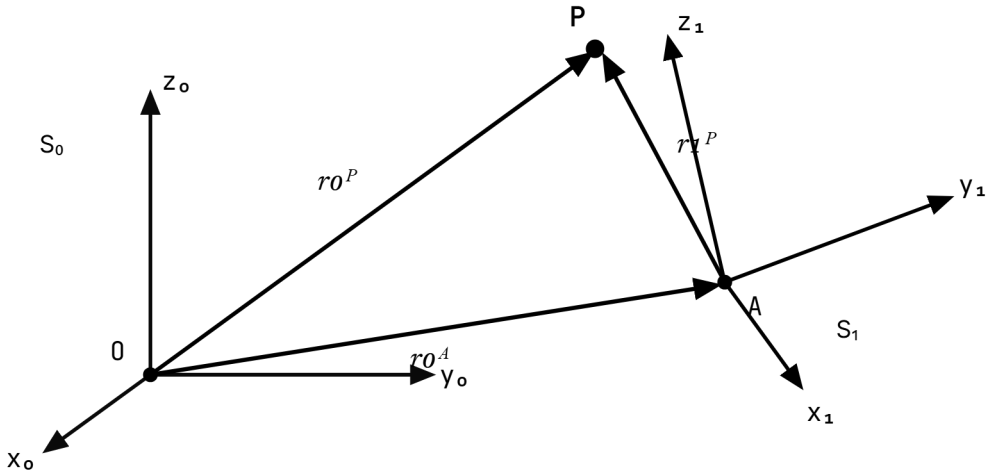
1. the **position of its origin** as a function of time, $\mathbf{r}_0^A(t)$;
2. the **attitude of its basis** as a function of time, $\mathbf{i}_1(t), \mathbf{j}_1(t), \mathbf{k}_1(t)$, for example through Euler angles $\psi(t), \theta(t), \varphi(t)$.

From these we compute $\mathbf{v}_0^A(t)$, $\mathbf{a}_0^A(t)$, $\boldsymbol{\omega}_{10}(t)$ and $\boldsymbol{\alpha}_{10}(t)$. The relative-motion expressions are especially useful when the motion of $S_1$ is **known a priori**: then none of these functions of time are unknowns (Slide 22, Notes Sec. 3.2).

## 10. Relative Kinematics of the Point Particle

### 10.1 Motivation

- We often have several observers and need to relate the position, velocity and acceleration of $P$ that each sees. Example: sensors on an experimental aircraft measure a hurricane; the motion of the aircraft must be taken into account to "translate" the measurements to a ground-based frame (Slide 23).
- In other cases it is simpler to obtain these vectors in a moving frame than in the fixed one (Slide 23).




*Fixed frame $S_0 = \{O; \mathcal{B}_0\}$ and moving frame $S_1 = \{A; \mathcal{B}_1\}$ observing the same point $P$ (Notes Fig. 3.2, Slide 23):* $\mathbf{r}_0^P = \mathbf{r}_0^A + \mathbf{r}_1^P$.

## 10.2 Position

$$\mathbf{r}_0^P = \mathbf{OP} = \mathbf{OA} + \mathbf{AP} = \mathbf{r}_0^A + \mathbf{r}_1^P$$

*Notes Eq. 3.16, Slide 23: absolute position = position of the origin of $S_1$ + relative position.*

## 10.3 Velocity

**Step 1.** The absolute velocity is the derivative of $\mathbf{r}_0^P$ in $S_0$; the derivative of a sum is the sum of the derivatives (Notes Eq. 3.17):

$$\mathbf{v}_0^P \equiv \left.\frac{d\mathbf{r}_0^P}{dt}\right|_0 = \underbrace{\left.\frac{d\mathbf{r}_0^A}{dt}\right|_0}_{\mathbf{v}_0^A} + \left.\frac{d\mathbf{r}_1^P}{dt}\right|_0$$

**Step 2.** $\mathbf{r}_1^P$ is naturally described in $S_1$, so apply the Coriolis theorem (Notes Eqs. 3.18–3.19), with $\left.d\mathbf{r}_1^P/dt\right|_1 \equiv \mathbf{v}_1^P$ the **relative velocity**:

$$\left.\frac{d\mathbf{r}_1^P}{dt}\right|_0 = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$$

$$\boxed{\mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P}$$

*Notes Eq. 3.20, Slide 24: velocity of the origin + relative velocity + term due to the rotation of $S_1$.*

## 10.4 Acceleration

**Step 1.** Differentiate the velocity in $S_0$ (Notes Eq. 3.21):

$$\mathbf{a}_0^P \equiv \left.\frac{d\mathbf{v}_0^P}{dt}\right|_0 = \underbrace{\left.\frac{d\mathbf{v}_0^A}{dt}\right|_0}_{\mathbf{a}_0^A} + \left.\frac{d\mathbf{v}_1^P}{dt}\right|_0 + \left.\frac{d}{dt}\right|_0 \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right)$$

**Step 2.** Coriolis on the relative velocity, with $d\mathbf{v}_1^P/dt\big|_1 \equiv \mathbf{a}_1^P$ the **relative acceleration** (Notes Eqs. 3.22 and 3.24):

$$\left.\frac{d\mathbf{v}_1^P}{dt}\right|_0 = \mathbf{a}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

**Step 3.** Product rule for the cross product (Notes Eq. 1.67), then Coriolis on $\mathbf{r}_1^P$ as in Section 10.3 (Notes Eq. 3.23):

$$\left.\frac{d}{dt}\right|_0 \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \underbrace{\left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0}_{\boldsymbol{\alpha}_{10}} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \underbrace{\left.\frac{d\mathbf{r}_1^P}{dt}\right|_0}_{\mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P} = \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right)$$

**Step 4.** Add the three terms. $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ appears twice (Steps 2 and 3), which gives the factor 2 (Notes Eq. 3.25):

$$\boxed{\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P}$$

| Term | Name (Slide 25) |
|---|---|
| $\mathbf{a}_0^A$ | acceleration of the origin of $S_1$ |
| $\mathbf{a}_1^P$ | relative acceleration |
| $\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ | Euler acceleration |
| $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$ | centripetal acceleration |
| $2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ | Coriolis acceleration |

## 10.5 Summary

$$\begin{aligned}\mathbf{r}_0^P &= \mathbf{r}_0^A + \mathbf{r}_1^P \\ \mathbf{v}_0^P &= \mathbf{v}_0^A + \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P \\ \mathbf{a}_0^P &= \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P\end{aligned}$$

*Slide 26, Notes Eqs. 3.26-3.27. The right-hand sides use only relative quantities plus the motion of $S_1$: $\mathbf{r}_0^A, \mathbf{v}_0^A, \mathbf{a}_0^A, \boldsymbol{\omega}_{10}, \boldsymbol{\alpha}_{10}$.*

## 10.6 Remarks on each term

1. The translation of $A$ appears only through $\mathbf{v}_0^A$ and $\mathbf{a}_0^A$; all other terms come from the relative motion of $P$ and/or the angular motion of $S_1$ (Notes p. 39).
2. The direction of $\boldsymbol{\omega}_{10}$ is the axis of rotation of $S_1$ relative to $S_0$.
3. $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$ is perpendicular to $\boldsymbol{\omega}_{10}$ and $\mathbf{r}_1^P$, with magnitude $\omega_{10}\, r_{1,\perp}^P$, where $r_{1,\perp}^P$ is the component of $\mathbf{r}_1^P$ perpendicular to $\boldsymbol{\omega}_{10}$.
4. The same holds for $\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ with respect to $\boldsymbol{\alpha}_{10}$; since $\boldsymbol{\alpha}_{10}$ and $\boldsymbol{\omega}_{10}$ are in general not parallel, it can have a component along $\boldsymbol{\omega}_{10}$.
5. The centripetal term has magnitude $r_{1,\perp}^P\, \omega_{10}^2$ and always points toward the line through the origin along $\boldsymbol{\omega}_{10}$.
6. The factor 2 of the Coriolis term comes from two effects of equal importance: the rotation of the relative velocity (Step 2) and the change of $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$ as $P$ moves (Step 3). It exists only in a rotating frame and only when $P$ has relative velocity.

## 10.7 Translating frame without rotation

$$\boldsymbol{\omega}_{10} = \boldsymbol{\alpha}_{10} = \mathbf{0}: \qquad \mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_1^P, \qquad \mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$$

*Notes Eqs. 3.28-3.29. If moreover $\mathbf{v}_0^A$ is constant ($A$ moves on a straight line), $\mathbf{a}_0^P = \mathbf{a}_1^P$: both observers measure the same acceleration.*

# 11. Relative Dynamics: Inertia Forces

CAN NEWTON'S LAW BE USED IN A MOVING FRAME?

Newton's laws are valid only in an inertial frame. Assuming $S_0$ is inertial, $m^P \mathbf{a}_0^P = \mathbf{F}$ (Notes Eq. 4.59). If $S_1$ accelerates or rotates relative to $S_0$ it is non-inertial. The relative acceleration formula shows how to fix Newton's law so that it can be applied in $S_1$ (Slide 27, Notes Sec. 4.9).

## 11.1 Derivation

**Step 1.** Substitute Notes Eq. 3.27 into Newton's law (Notes Eq. 4.60):

$$m^P \left[\mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P\right] = \mathbf{F}$$

**Step 2.** Keep the relative acceleration on the left (Notes Eq. 4.61):

$$m^P \mathbf{a}_1^P = \mathbf{F} - m^P \mathbf{a}_0^A - m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m^P \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) - 2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

**Step 3.** Define the **fictitious** or **inertia force** (Notes Eq. 4.62):

$$\mathbf{F}_i = -m^P \mathbf{a}_0^A - m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m^P \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) - 2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

$$\boxed{m^P \mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i}$$

*Notes Eq. 4.63, Slide 27.*

Eq. 4.63 contains the same information as Eq. 4.60: terms have only been moved across. The new reading is that an observer in $S_1$ may apply Newton's second law if the force $\mathbf{F}_i$ is added to the actual forces. Every term of $\mathbf{F}_i$ is computed from the relative position and velocity of $P$ once $\mathbf{a}_0^A, \boldsymbol{\omega}_{10}, \boldsymbol{\alpha}_{10}$ are known (Notes p. 53).

## 11.2 Names of the four terms

| Term | Name (Slide 28, Notes p. 53) | Symbol |
|---|---|---|
| $-m^P \mathbf{a}_0^A$ | force due to the acceleration of the origin | $\mathbf{F}_A$ |
| $-m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ | Euler's force | $\mathbf{F}_E$ |
| $-m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$ | centrifugal force due to the rotation of $S_1$ | $\mathbf{F}_C$ |
| $-2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ | Coriolis force | $\mathbf{F}_{COR}$ |

### 11.3 General properties (Slide 28)

- All inertia forces are **proportional to the mass** of the particle.
- Forces 1 to 3 are **locally indistinguishable from a gravitational field**: an observer in $S_1$ cannot tell them apart from actual gravity.
- Force 4 (Coriolis) is different: it is zero when $P$ is at rest in $S_1$ and appears only when $P$ has relative velocity with a component perpendicular to $\boldsymbol{\omega}_{10}$.
- We can obtain the equations of motion in inertial and non-inertial frames; **choose the simplest option for each problem**.

### 11.4 Physical intuition (Notes p. 53)

Although fictitious, these forces may feel very real to an observer on a non-inertial frame; they are due solely to the motion of that frame relative to an inertial one.

- **Braking vehicle.** We tend to go forward when a vehicle brakes, and it is common to say the inertia force $-m\mathbf{a}_0^A$ pushes us. For a ground observer no such force exists: the vehicle decelerates through the friction of the wheels and the passenger's mass tends to continue forward.
- **Turning vehicle.** The road gives the vehicle a transverse friction force toward the centre of curvature. The passenger tends to continue in uniform rectilinear motion and feels pushed outward, until the seat and the floor transmit the necessary force.
- **Accelerating merry-go-round.** Besides the outward centrifugal force, a child feels a force pointing backward with respect to the direction of travel: the Euler term $-m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$.
- **Ball on the merry-go-round.** A ball passed from the outer to the inner part turns away from the expected straight line toward the direction of travel: the Coriolis force. It also explains the sense of rotation of hurricanes in each hemisphere, and why a cannonball shot toward the East lands farther than one shot equally toward the West.

## 12. Energy and Potentials in a Non-Inertial Frame

### 12.1 Energy and other quantities in $S_1$

The kinetic energy $T_1^P = \frac{1}{2} m^P \, \mathbf{v}_1^P \cdot \mathbf{v}_1^P$ and the mechanical energy $E_1^P$ of $P$ can always be defined in $S_1$, even if non-inertial. If the necessary conditions are met (the same as before, with the inertia forces counted among the forces), $E_1^P$ is a conserved quantity. The angular momentum $\mathbf{H}_{A1}^P$ or any other quantity used so far can also be defined in $S_1$ (Slide 29).

### 12.2 Force due to the acceleration of the origin

$$\mathbf{a}_0^A = \text{const}: \qquad V_1 = m^P \, \mathbf{a}_0^A \cdot \mathbf{r}_1^P, \qquad \mathbf{F}_A = -\nabla V_1 = -m^P \mathbf{a}_0^A$$

*Slide 29. With $V_1 = m^P(a_x x_1 + a_y y_1 + a_z z_1)$, $\partial V_1/\partial x_1 = m^P a_x$, etc. $\mathbf{F}_A$ is a homogeneous field in $S_1$, like uniform gravity, pointing opposite to $\mathbf{a}_0^A$.*

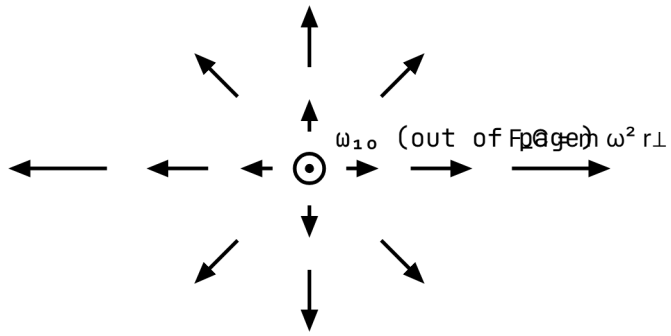
## 12.3 Centrifugal force

Decompose $\mathbf{r}_1^P = \mathbf{r}_\parallel + \mathbf{r}_\perp$ along and across the axis, with $\boldsymbol{\omega}_{10} = \omega_{10}\mathbf{e}_\omega$ and $\mathbf{r}_\parallel = r_\parallel \mathbf{e}_\omega$ (Slide 30). Using $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ and $\boldsymbol{\omega}_{10}\cdot\mathbf{r}_1^P = \omega_{10} r_\parallel$:

$$\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = \left(\boldsymbol{\omega}_{10}\cdot\mathbf{r}_1^P\right)\boldsymbol{\omega}_{10} - \left(\omega_{10}\right)^2\mathbf{r}_1^P = \left(\omega_{10}\right)^2\left[r_\parallel\mathbf{e}_\omega - \mathbf{r}_1^P\right] = -\left(\omega_{10}\right)^2\mathbf{r}_\perp$$

$$\mathbf{F}_C = m^P\left(\omega_{10}\right)^2\mathbf{r}_\perp, \qquad \boldsymbol{\omega}_{10} = \text{const}: \quad V_1 = -\frac{1}{2}m^P\left(\omega_{10}\right)^2\left(r_\perp\right)^2$$

*Slide 30. Check:* $(r_\perp)^2 = \mathbf{r}_1^P\cdot\mathbf{r}_1^P - (\mathbf{e}_\omega\cdot\mathbf{r}_1^P)^2$, *so* $\nabla(r_\perp)^2 = 2\mathbf{r}_1^P - 2(\mathbf{e}_\omega\cdot\mathbf{r}_1^P)\mathbf{e}_\omega = 2\mathbf{r}_\perp$ *and* $-\nabla V_1 = m^P\omega_{10}^2\mathbf{r}_\perp$.




*Centrifugal force field in $S_1$ seen along $\boldsymbol{\omega}_{10}$ (Slide 30): radial, outward from the axis, proportional to $r_\perp$.*

## 12.4 Euler force

$\mathbf{F}_E = -m^P\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P$ is always perpendicular to $\boldsymbol{\alpha}_{10}$ and $\mathbf{r}_1^P$, and it does **not** admit a potential: $\nabla\times\mathbf{F}_E \neq \mathbf{0}$ (Slide 31). For a given $\boldsymbol{\alpha}_{10}$, $\nabla\times(\boldsymbol{\alpha}_{10}\times\mathbf{r}) = 2\boldsymbol{\alpha}_{10}$, so $\nabla\times\mathbf{F}_E = -2m^P\boldsymbol{\alpha}_{10}$.

## 12.5 Coriolis force

$\mathbf{F}_{COR} = -2m^P\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ is always perpendicular to $\boldsymbol{\omega}_{10}$ and $\mathbf{v}_1^P$, and proportional to $v_\perp$, the component of $\mathbf{v}_1^P$ perpendicular to $\boldsymbol{\omega}_{10}$ (Slide 31). Its power vanishes:

$$\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = -2m^P\left(\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P\right)\cdot\mathbf{v}_1^P = 0$$

$\mathbf{F}_{COR}$ *never exerts work (Slide 31), so it never spoils the conservation of* $E_1^P$.

**Example (Slide 31): equatorial launch**

Slide 31

$\boldsymbol{\omega}_{10} = \Omega\,\mathbf{e}_N$ is horizontal at the equator

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1. Local right-handed unit vectors East $\mathbf{e}_E$, North $\mathbf{e}_N$, Up $\mathbf{e}_U$: $\mathbf{e}_N \times \mathbf{e}_E = -\mathbf{e}_U$.
2. **Toward the East**, $\mathbf{v}_1^P$ has a component $v\,\mathbf{e}_E$: $-2m\Omega\,\mathbf{e}_N \times (v\,\mathbf{e}_E) = 2m\Omega v\,\mathbf{e}_U$, **upward**. The trajectory is higher and longer.
3. **Toward the West** ($v\,\mathbf{e}_E$ with $v < 0$) the force points **downward**: lower and shorter trajectory. This is the cannonball effect of Section 11.4.

### 12.6 Conservation of $E_1^P$ in a frame rotating at constant rate

With a fixed origin ($\mathbf{a}_0^A = \mathbf{0}$) and constant $\boldsymbol{\omega}_{10}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$), the Euler force vanishes, the centrifugal force has a potential and the Coriolis force does no work. If the actual forces that work are conservative and the constraints are fixed in $S_1$:

$$E_1^P = \frac{1}{2}m^P \left|\mathbf{v}_1^P\right|^2 + V(\mathbf{r}_1^P) - \frac{1}{2}m^P\omega_{10}^2 r_\perp^2 = \text{const}$$

*The energy integral of the rotating-wire and rotating-ring problems. If $\boldsymbol{\omega}_{10}$ depends on time, the Euler force appears and $E_1^P$ is in general not conserved.*

## 13. Method for Relative-Motion Problems

This checklist condenses Slides 22, 26 and 28 into the order used in the solved problems:

1. **Choose the frames.** $S_0$ inertial; $S_1$ attached to the moving body (wire, ring, platform, spacecraft) so that the constraint or the motion is simple in $S_1$.
2. **Describe the motion of $S_1$:** $\mathbf{r}_0^A, \mathbf{v}_0^A, \mathbf{a}_0^A, [{}_0R_1], \boldsymbol{\omega}_{10}, \boldsymbol{\alpha}_{10}$ (Section 9).
3. **Relative kinematics of $P$:** $\mathbf{r}_1^P, \mathbf{v}_1^P, \mathbf{a}_1^P$ with the coordinates of the problem.
4. **Either** compute $\mathbf{a}_0^P$ with Notes Eq. 3.27 and write $m\mathbf{a}_0^P = \mathbf{F}$ in $S_0$, **or** write $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ in $S_1$ (Notes Eq. 4.63). Both give the same equations.
5. **Project** on the directions where the reactions vanish to get the equations of motion; project on the others to get the reactions.
6. **First integrals:** $E_1^P$ if $\mathbf{a}_0^A$ and $\boldsymbol{\omega}_{10}$ are constant (or zero) and the constraints are fixed in $S_1$.

## 14. Source Map

| Content | Notes.pdf (printed pages) | Slides |
|---|---|---|
| Observers, absolute and relative | Ch. 3 introduction (p. 36) | 3, 23 |
| Translational motion of a frame | Sec. 3.2 (p. 38) | 4 |

| Content | Notes.pdf (printed pages) | Slides |
|---|---|---|
| Change of basis, properties, composition | Sec. 1.5, Eqs. 1.6–1.17 (pp. 13–14) | 5–7 |
| Axis and angle of rotation | Sec. 1.5.1 (pp. 14–15) | 6 |
| Simple rotations | Sec. 1.5.2, Eqs. 1.18–1.20 (pp. 15–16) | 8 |
| Euler angles, 3-2-1 example | Sec. 1.5.3, Eqs. 1.21–1.23 (pp. 16–17) | 9–13 |
| Angular velocity and acceleration | Sec. 1.7, Eqs. 1.62, 1.73; Eqs. 3.3–3.5, 3.9 (pp. 21–25, 37) | 14–16 |
| Coriolis theorem | Sec. 3.1, Eqs. 3.1–3.8 (pp. 36–37) | 17–18 |
| Composition of angular motions | Eqs. 3.10–3.15 (pp. 37–38) | 19 |
| Angular velocity from Euler angles | none | 20–21 |
| Configuration and motion of a frame | Sec. 3.2 (p. 38) | 22 |
| Relative kinematics | Sec. 3.2, Eqs. 3.16–3.29 (pp. 38–40) | 23–26 |
| Relative dynamics, inertia forces | Sec. 4.9, Eqs. 4.59–4.63 (pp. 52–53) | 27–28 |
| Potentials, Euler and Coriolis forces | none | 29–31 |

Worked problems (Problems 7, 9, 20, 21, 25, 26, 27, 28, 30, 33, 34, 36, 37, 40, 41, 47, 49, 51) follow the four-phase protocol on the Topic 5 problems page.